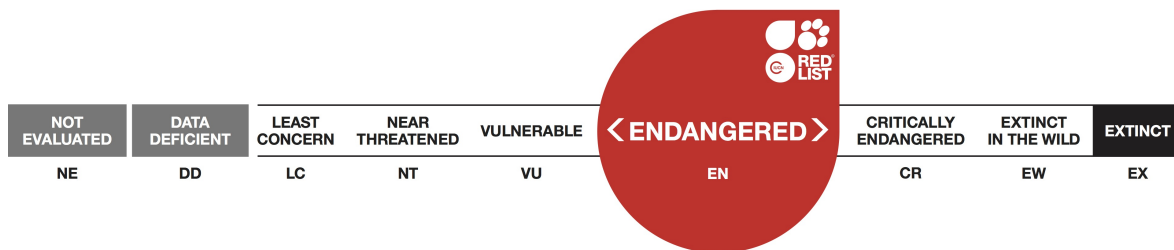


Rungwecebus kipunji, Kipunji

Assessment by: Davenport, T.



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Mammalia	Primates	Cercopithecidae

Taxon Name: *Rungwecebus kipunji* (Ehardt, Butynski, Jones & Davenport, 2005)

Synonym(s):

- *Lophocebus kipunji* (Ehardt, Butynski, Jones & Davenport, 2005)

Common Name(s):

- English: Kipunji
- French: Kipunji
- Spanish: Kipunji

Taxonomic Source(s):

Mittermeier, R.A., Rylands, A.B. and Wilson D.E. 2013. *Handbook of the Mammals of the World: Volume 3 Primates*. Lynx Edicions, Barcelona.

Taxonomic Notes:

Africa's most recently discovered (2003) and described (2006) genus of monkey (Davenport *et al.* 2006). Originally allocated to the genus *Lophocebus* (Jones *et al.* 2005), but later placed in the monotypic genus *Rungwecebus* by Davenport *et al.* (2006) on the basis of molecular and morphological data. The genus to which *R. kipunji* belongs was questioned by Ehardt and Butynski (2006), but considerable further evidence supports the phylogenetic position and taxonomic status of *Rungwecebus* (Olson *et al.* 2008, Roberts *et al.* 2009). The evolutionary implications of *Rungwecebus* being a sister taxon to *Papio* are considerable (Olson *et al.* 2008, Davenport 2013).

Assessment Information

Red List Category & Criteria: Endangered B1ab(ii,iii,iv,v)+2ab(ii,iii,iv,v) [ver 3.1](#)

Year Published: 2019

Date Assessed: March 20, 2018

Justification:

Rungwecebus kipunji is sparsely distributed (Davenport *et al.* 2006, 2008; Bracebridge *et al.* 2011, 2012). The small areas of habitat where it occurs (42.4 km² in Rungwe-Kitulo and ~25 km² in Ndundulu), and the species' patchy distribution in Rungwe-Kitulo give grounds for considerable conservation concern. Totalling 68 km², its area of occupancy (AOO) is below the 500 km² upper threshold for an Endangered listing under criterion B2. At 3,518 km², the taxon's extent of occurrence (EOO) also meets the threshold for criterion B1. In addition, the habitat is fragmented and remote sensing analyses of forest loss in Rungwe-Kitulo indicate a decline over the last 30 years, which although now slowing, is continuing. The management of Mt Rungwe is currently dependent on funds from the Wildlife Conservation Society (WCS). Forest loss has been reduced but continues in key areas. Persistent hunting of these monkeys across Rungwe-Kitulo (and possibly Ndundulu) – especially individuals which raid crops – as well as the

precarious nature of the small population in Ndundulu, where logging, charcoal making and unmanaged resource extraction are commonplace, lead us to project a continuing decline in AOO, area, extent and quality of habitat, and the number of mature individuals remaining. As a consequence, the species is here listed as EN B1ab+2ab.

Previously Published Red List Assessments

2008 – Critically Endangered (CR)

<http://dx.doi.org/10.2305/IUCN.UK.2008.RLTS.T136791A4340286.en>

Geographic Range

Range Description:

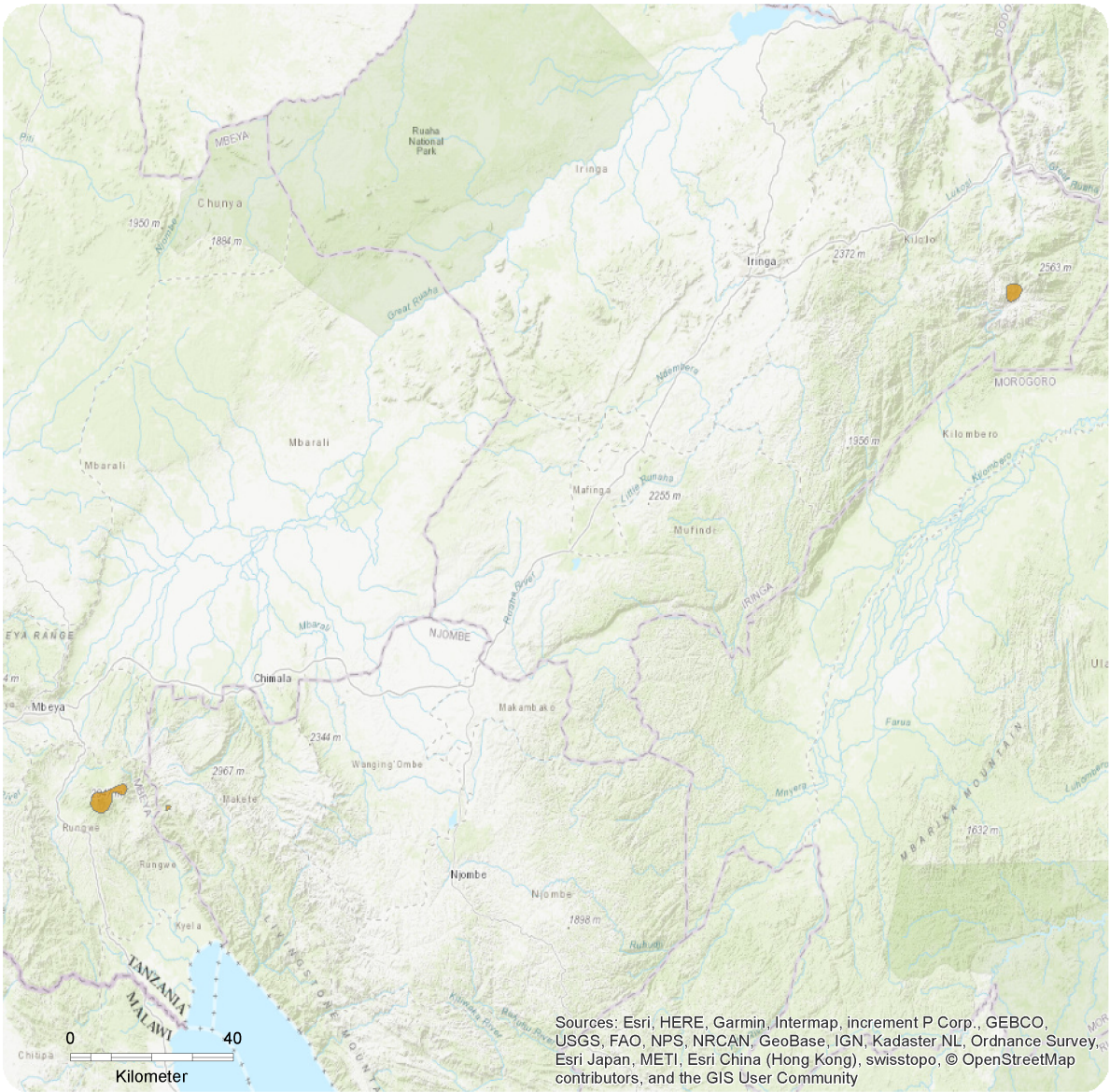
A Tanzanian endemic known from two populations separated by *ca* 350 km of non-forested, agricultural land. One population is at 1,750–2,450 m asl in 12.4 km² of Rungwe-Kitulo Forest (Davenport *et al.* 2008) in the Southern Highlands of south-western Tanzania (09.12°–09.18°S, 33.67°–33.92°E). The Rungwe-Kitulo Forest includes Mt Rungwe Nature Reserve (150 km²) and the Livingstone Forest (191 km²), which lies within the 412 km² Kitulo National Park (Davenport 2002, Davenport and Bytebier 2004, Davenport *et al.* 2005, 2006, 2008). Mt Rungwe and Livingstone Forest (in Kitulo) are connected by the Bujingijila Corridor, a 2-km wide degraded forest connection (Davenport 2005, 2006, Mwakilema and Davenport 2005). The Kipunji inhabits the wetter forest of south Mt Rungwe and isolated groups are scattered in the north and south of Livingstone Forest (Davenport *et al.* 2006, 2008, Bracebridge *et al.* 2011). The other population of *R. kipunji* is found at 1,300–1,750 m asl in the Vikongwa Valley, Ndundulu Forest (07.67°–07.85°S, 35.17°–36.83°E; *ca* 180 km of closed forest), in the Kilombero Nature Reserve in the Udzungwa Mountains, south-central Tanzania (Jones *et al.* 2015). The area known to be occupied by the species in the Ndundulu Forest is approximately 25 km² (Davenport pers obs. in Marshall *et al.* 2015) and Kipunji density is low. Research indicates that the animal is absent from the contiguous Udzungwa Mountains National Park (Davenport *et al.* 2008, Marshall *et al.* 2015). The species' area of occupancy (AOO) is estimated to be 42.4 km² in Rungwe-Kitulo (Bracebridge *et al.* 2011) and ~25 km² in Ndundulu (Davenport pers. obs. in Marshall *et al.* 2015), with a combined total AOO of only 67.5 km².

Country Occurrence:

Native: Tanzania, United Republic of

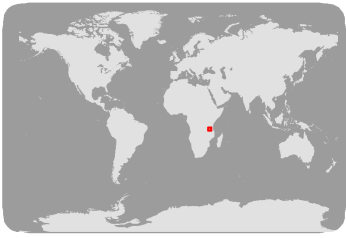
Distribution Map

Rungwecebus kipunji



Range
Extant (resident)

Compiled by:
IUCN (International Union for Conservation of Nature)



Population

Surveys totalling 2,864 hours and covering 3,456 km of transect were undertaken by Davenport *et al.* (2008) to determine distribution and group numbers. In addition, 772 hours of simultaneous multi-group observational follows in Rungwe-Kitulo and Ndundulu forests enabled 209 total counts to be carried out. Davenport *et al.* (2008) estimated some 1,042 individuals in Rungwe-Kitulo, ranging from 25 to 39 individuals per group ($\mu = 30.65$; SE = 0.62; $n = 34$), and 75 individuals in Ndundulu, ranging from 15 to 25 individuals per group ($\mu = 18.75$; SE = 2.39; $n = 4$). The total Kipunji population was thus estimated to be 1,117 animals in 38 groups ($\mu = 29.39$; SE = 0.85; $n = 38$). The Ndundulu population is restricted and the Rungwe-Kitulo population is highly fragmented with isolated subpopulations remaining in degraded habitat (Davenport *et al.* 2008). Kipunji densities appear to be over three times greater in the secondary forests of Rungwe-Kitulo than in the primary forests of the Udzungwa population (Davenport *et al.* 2008), which is likely an effect of forest quality, but also perhaps competition with other diurnal primates not present in Rungwe-Livingstone (Bracebridge *et al.* 2011).

Current Population Trend: Decreasing

Habitat and Ecology (see Appendix for additional information)

Habitat ranges from degraded montane and upper montane forest from 1,700–2,150 m in Rungwe-Kitulo (Davenport *et al.* 2006, 2008, Bracebridge *et al.* 2011, 2012) to submontane forest in Ndundulu (Davenport and Jones 2005, Jones *et al.* 2005). In Rungwe-Kitulo, the canopy is often broken and between 10 and 30 m with emergents reaching 35 m. Kipunji prefers steep-sided gullies and valley edges and is rarely found far from streams (Davenport 2013). Ridges and open areas are generally avoided and the animals are not territorial (De Luca *et al.* 2010). The forest was much reduced by logging from 1960s to 1980s (Lovett 1986, McKone and Walzem 1994, Machaga *et al.* 2004, Davenport 2005, 2006). Thick undergrowth is typical, with the tree fern *Cyathea manniana*, wild banana *Ensete ventricosum*, and large stands of bamboo *Sinarundinaria alpina* common in the south and southeast of Mt Rungwe and the northwest of Livingstone. The upper limits of this distribution are likely due to the floristic changes at around 2,000–2,200m asl, where bamboo *Sinarundinaria alpina*, often interspersed with *Hagenia abyssinica* or *Podocarpus latifolius*, dominate (Bracebridge *et al.* 2011). The species rarely frequents the bamboo and leaves the forest only to raid nearby crops, especially maize. In southern Mt Rungwe, annual rainfall ranges from ca 185–280 cm and there is an indistinct and short dry season from June to October. In Ndundulu, the *Parinari excelsa*-dominant forest is undisturbed and the canopy is unbroken, reaching heights of 40–50 m. When comparing predictions from an interpolation analysis for a habitat suitability model with the EOO, the vast majority of the most suitable areas in Rungwe-Kitulo were already occupied by Kipunji (Bracebridge *et al.* 2011). The best forest for Kipunji would be at mid to lower altitude with a closed canopy. However, as much of Rungwe-Kitulo contains heterogeneous forest, reflecting a history of disturbance (Davenport *et al.* 2006, Davenport, 2009), forest is often fragmented and contains a mix of primary, secondary and regenerating forest. This suggests that while not optimal, Kipunji can tolerate disturbed habitat reasonably well (Bracebridge *et al.* 2011).

Systems: Terrestrial

Use and Trade

Occasionally hunted for food. More often hunted as retribution for crop-raiding using log traps or dogs (Davenport 2005, 2006). Hunting of Kipunji is illegal.

Threats (see Appendix for additional information)

The Rungwe-Kitulo forests are degraded. Logging, charcoal making, illegal hunting and unmanaged resource extraction have long been commonplace (Machaga *et al.* 2004, Davenport 2005, 2006). The narrow Bujingijila corridor linking Mt Rungwe Nature Reserve to Livingstone Forest in Kitulo National Park, and the corridors joining the northern and southern sections of Livingstone, are encroached by farmers and degraded (Davenport 2005, Mwakilema and Davenport 2005). Subpopulations are therefore fragmented even within the same forest protected area. Kipunji is hunted with log traps and with also with dogs (Davenport 2005, 2006), mainly as retribution for the crop-raiding of maize (and the leaves of beans and potatoes) and primarily from January to April. However, the monkey is also occasionally hunted for food. Although Ndundulu Forest is in good condition and largely undisturbed, the species is present there in low numbers (Davenport *et al.* 2008) and the reasons for this are unclear (Marshall *et al.* 2015). Both Kilombero and Mt Rungwe Nature Reserves lack management resources.

Conservation Actions (see Appendix for additional information)

Rungwecebus kipunji is listed on Appendix II of CITES. Kipunji occurs entirely in protected areas, 48.4% in Kitulo National Park, 44.9% in Mt Rungwe Nature Reserve and 6.7% in the Kilombero Nature Reserve (Davenport *et al.* 2008). However, only the national park has independent management activities and resources, and they remain limited, especially in the forest. The focus of current Kipunji conservation work is the protection and restoration of the montane habitats of Mt Rungwe, especially the forest connections, such as the (2.1 km²) Bujingijila corridor (Davenport 2006, Bracebridge *et al.* 2011). Protecting corridors is of the highest priority for the conservation of this genus/species and Bujingijila could provide habitat for an additional 88 Kipunji (8% population increase), using density estimates from the 2008 census (Bracebridge *et al.* 2013). It also has the benefit of reconnecting the Mt Rungwe and Kitulo kipunji subpopulations. Furthermore, conservation management, while focusing on promoting old growth forest, should also consider populations of some important pioneer tree species such as *Macaranga capensis*, both in the forest and in reforestation schemes outside the species' current Area of Occupancy (Bracebridge *et al.* 2012).

Kipunji is being used as a 'flagship species' by WCS's long-term Southern Highlands Conservation Program in and around Rungwe-Kitulo, especially in education and awareness raising activities, and as part of a long-term monitoring programme. A habituated group is monitored daily for research, and a new group is being habituated for tourism.

Credits

Assessor(s): Davenport, T.

Reviewer(s): Williamson, E.A.

Contributor(s): Jones, T.

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Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
1. Forest -> 1.9. Forest - Subtropical/Tropical Moist Montane	Resident	Suitable	Yes
14. Artificial/Terrestrial -> 14.4. Artificial/Terrestrial - Rural Gardens	Passage	Marginal	-
14. Artificial/Terrestrial -> 14.6. Artificial/Terrestrial - Subtropical/Tropical Heavily Degraded Former Forest	Resident	Marginal	-

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity	Impact Score
1. Residential & commercial development -> 1.3. Tourism & recreation areas	Ongoing	Minority (50%)	Negligible declines	Low impact: 4
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.2. Species disturbance 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.5. Inbreeding 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		
11. Climate change & severe weather -> 11.1. Habitat shifting & alteration	Ongoing	Majority (50-90%)	Slow, significant declines	Medium impact: 6
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation		
2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.1. Shifting agriculture	Ongoing	Minority (50%)	Causing/could cause fluctuations	Low impact: 5
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		
2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.2. Small-holder farming	Ongoing	Minority (50%)	Causing/could cause fluctuations	Low impact: 5
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		
5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.1. Intentional use (species is the target)	Ongoing	Minority (50%)	Rapid declines	Medium impact: 6
	Stresses:	2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.2. Species disturbance 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		

5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.2. Unintentional effects (species is not the target)	Ongoing	Minority (50%)	Slow, significant declines	Low impact: 5
	Stresses:	2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.2. Species disturbance 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		
5. Biological resource use -> 5.1. Hunting & trapping terrestrial animals -> 5.1.3. Persecution/control	Ongoing	Minority (50%)	Slow, significant declines	Low impact: 5
	Stresses:	2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.2. Species disturbance 2. Species Stresses -> 2.3. Indirect species effects -> 2.3.7. Reduced reproductive success		
5. Biological resource use -> 5.3. Logging & wood harvesting -> 5.3.3. Unintentional effects: (subsistence/small scale) [harvest]	Ongoing	Minority (50%)	Slow, significant declines	Low impact: 5
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.2. Species disturbance		

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Actions in Place
In-Place Research, Monitoring and Planning
Systematic monitoring scheme: Yes
In-Place Land/Water Protection and Management
Conservation sites identified: Yes, over entire range
Occur in at least one PA: Yes
In-Place Education
Subject to recent education and awareness programmes: Yes
Included in international legislation: Yes
Subject to any international management/trade controls: Yes

Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Actions Needed
1. Land/water protection -> 1.1. Site/area protection
1. Land/water protection -> 1.2. Resource & habitat protection
2. Land/water management -> 2.1. Site/area management

Conservation Actions Needed
2. Land/water management -> 2.2. Invasive/problematic species control
2. Land/water management -> 2.3. Habitat & natural process restoration
3. Species management -> 3.2. Species recovery
4. Education & awareness -> 4.1. Formal education
4. Education & awareness -> 4.2. Training
4. Education & awareness -> 4.3. Awareness & communications
5. Law & policy -> 5.1. Legislation -> 5.1.2. National level
5. Law & policy -> 5.3. Private sector standards & codes
5. Law & policy -> 5.4. Compliance and enforcement -> 5.4.2. National level
6. Livelihood, economic & other incentives -> 6.1. Linked enterprises & livelihood alternatives
6. Livelihood, economic & other incentives -> 6.4. Conservation payments

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed
2. Conservation Planning -> 2.1. Species Action/Recovery Plan
2. Conservation Planning -> 2.2. Area-based Management Plan
3. Monitoring -> 3.1. Population trends
3. Monitoring -> 3.4. Habitat trends

Additional Data Fields

Distribution
Estimated area of occupancy (AOO) (km ²): 67.5
Continuing decline in area of occupancy (AOO): Yes
Extreme fluctuations in area of occupancy (AOO): No
Estimated extent of occurrence (EOO) (km ²): 3518
Number of Locations: 2
Continuing decline in number of locations: Yes
Extreme fluctuations in the number of locations: No
Lower elevation limit (m): 1300
Upper elevation limit (m): 2450

Population
Continuing decline of mature individuals: Yes
Extreme fluctuations: No
Continuing decline in subpopulations: No
Extreme fluctuations in subpopulations: No
All individuals in one subpopulation: No
Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Yes
Movement patterns: Not a Migrant

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